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DESIGN PORTFOLIO

Aryaman Bhatnagar

CAD Designs · CFD & FEA Simulations

PORTFOLIO SUMMARY

Electric Motorcycle	SOLIDWORKS · Featured on DS
Sports Motorcycle	Siemens NX · Sub-D surfacing
Motorcycle CFD	Ansys Fluent · Thesis study
F1 Car 2026	SOLIDWORKS + CFD simulation
F1 Car 2022	SOLIDWORKS · Ground effect
F1 Car 2021	PTC Creo · First F1 project
Sedan FEA + CFD	Ansys Static + Fluent 2024
Hypercar Concept	Fusion 360 T-Spline

SOLIDWORKS and CATIA | Siemens NX | PTC Creo | Fusion 360

Ansys Fluent | Ansys Mechanical | OPENFOAM | ALTAIR Hypermesh

SKILLS & SOFTWARE EXPERTISE

SOLIDWORKS & CATIA

CAD / FEA

Part Design · Weldments · Sheet Metal · Surfacing · Generative Shape Design (GSD)

Siemens NX

Advanced CAD

Pipe Routing · Freeform Surface Modelling · Solid Modelling· Assembly · Additive Manufacturing

PTC Creo and Fusion 360

Generative CAD

Freestyle / Advanced Surfacing ·T-Spline Modelling · Rendering · CAM

Ansys Fluent

CFD

Turbulence Modelling · Contours · Postprocessing (Pressure, Velocity, Turbulence, Vorticity) · Mesh Independence Study

Ansys Mechanical

FEA

Static Structural · Deformation · Stress Analysis Explicit Dynamics· Topology Optimisation · Modal Analysis

Engineering Drawing

Manufacturing Drawing

Assembly Drawing

Technical Drawings · Tolerance Standards · GD&T

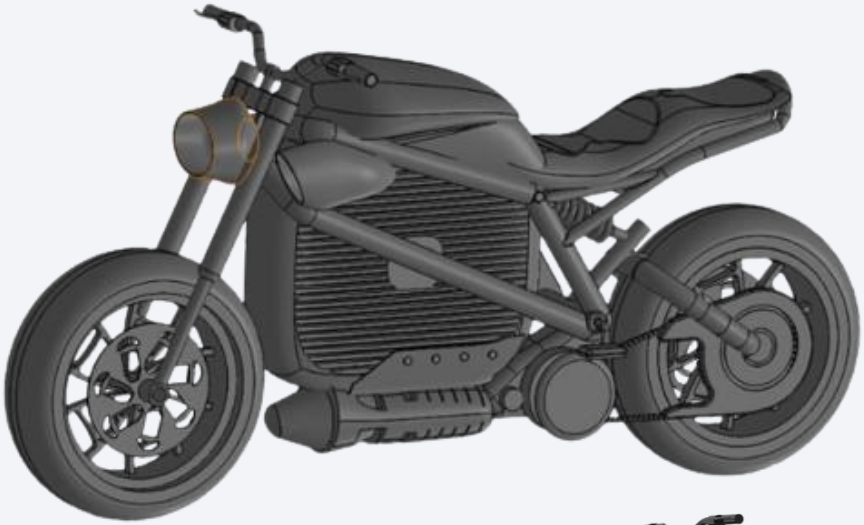
01

MOTORCYCLE DESIGN

ELECTRIC MOTORCYCLE

PROJECT OVERVIEW

A full-scale electric motorcycle designed from the ground up in SOLIDWORKS. The design was recognised and featured on the official Dassault Systèmes LinkedIn and Instagram pages, highlighting its technical depth and visual quality.



KEY FEATURES & HIGHLIGHTS

1. Frame developed using SOLIDWORKS Weldments with parametric structural profiles and trimmed joint intersections
2. Sheet metal body panels designed with defined K-factor and bend allowance for manufacturability
3. Full assembly comprising 50+ components with constrained mating strategy
4. Surface modelling applied to storage tank, seat, and wheel geometry
5. Photorealistic rendering using KeyShot for design visualisation

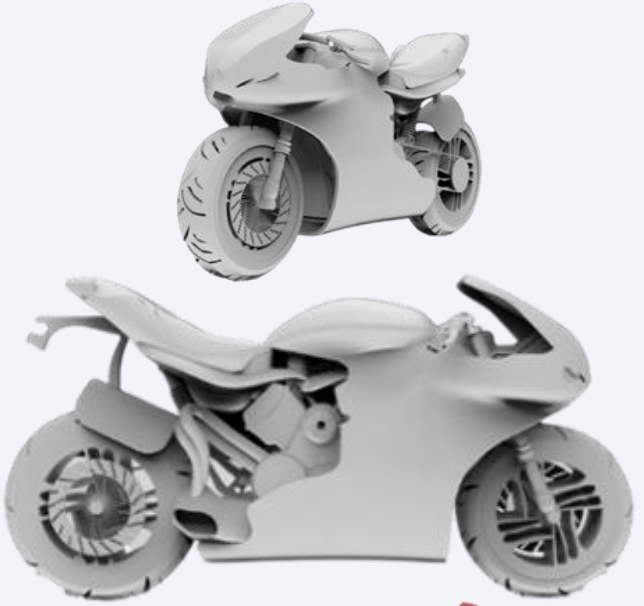
TECHNICAL SPECIFICATIONS

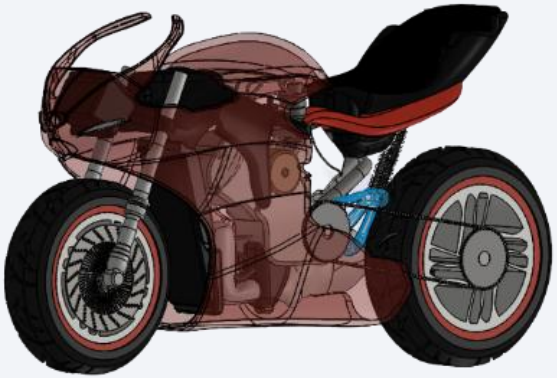
CAD Platform	SOLIDWORKS 2024
Frame Method	Weldments (structural profiles)
Body Method	Sheet Metal (bend-controlled geometry)
Components	Part Design + Assembly
Rendering	KeyShot
Recognition	Dassault Systèmes featured

SPORTS MOTORCYCLE

PROJECT OVERVIEW

A high-performance sports motorcycle modelled in Siemens NX, focusing on advanced surface modelling and geometric control of complex aerodynamic forms.





KEY FEATURES & HIGHLIGHTS

1. Sub-D (3D Form) modelling used to generate smooth, continuous aerodynamic fairing surfaces
2. Pipe routing tools applied to define frame geometry with controlled bend radii
3. Detailed component modelling including drivetrain, suspension, braking system, and wheels
4. Rendered in multiple angles with metallic paint & procedural textures

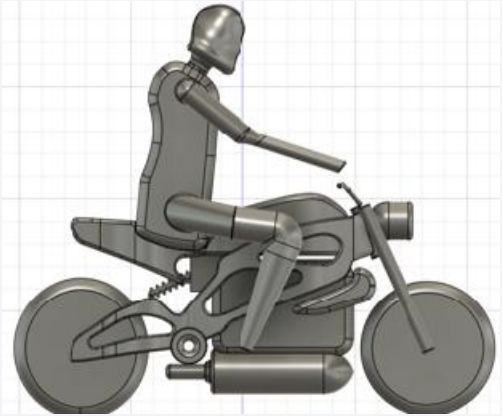
TECHNICAL SPECIFICATIONS

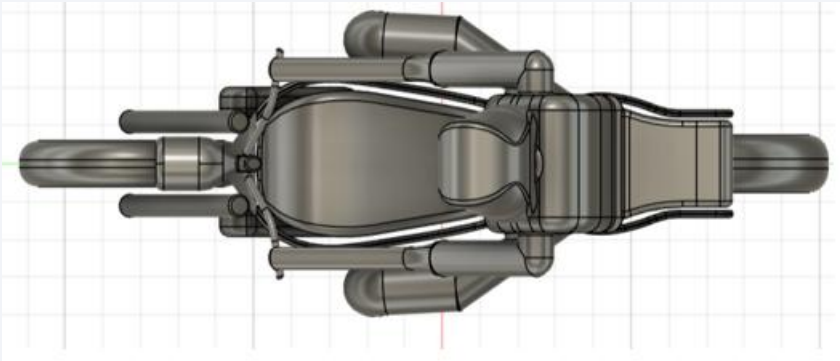
CAD Platform	Siemens NX
Frame Method	Pipe Routing (bend radius control)
Body Method	3D Form Sub-D surface modelling
Surface Quality	High-quality Sub-D surfaces with curvature continuity
Assembly	Full system integration
Rendering	NX Render + post-processing

DISSERTATION: NAKED TYPE MOTORCYCLE

PROJECT OVERVIEW

A detailed motorcycle geometry model developed in SOLIDWORKS, focused on creating an aerodynamically representative naked-type motorcycle suitable for downstream simulation and design evaluation. The project emphasises parametric modelling, surface control, and assembly integration.





KEY FEATURES & HIGHLIGHTS

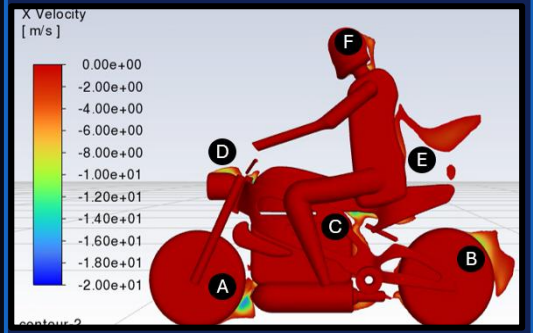
1. Parametric CAD modelling approach used to define all major components and ensure design flexibility
2. Frame, wheels, and mechanical components created using Extrude, Revolve, and Sweep features
3. Fuel tank, seat, and aerodynamic body regions developed using Lofted Surfaces, Boundary Surfaces, and Fillets to ensure smooth transitions
4. Full motorcycle assembly created with multiple mated components

TECHNICAL SPECIFICATIONS

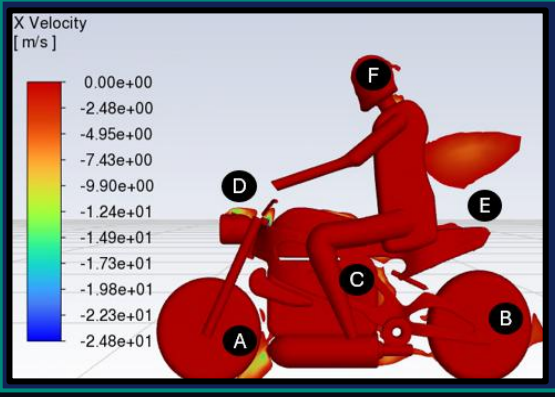
CAD Platform	SOLIDWORKS
Frame Method	Parametric CAD
Body Method	Lofted and Boundary surfaces
Assembly	Full mated components
Meshing	Body of Influence

MSc Dissertation CFD Results

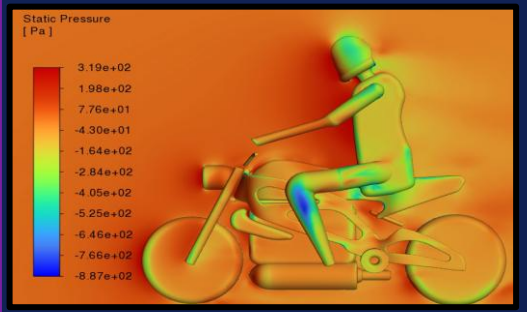
X-Velocity Contour (Baseline)



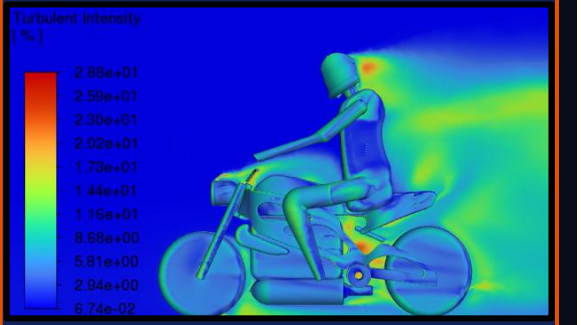
X-Velocity Contour (With VGs)



Static Pressure [Pa]



Turbulent Intensity (%)



Using CFD To Investigate The Effect Of Triangular Vortex Generators On The Aerodynamic Performance Of A Naked-type Motorcycle

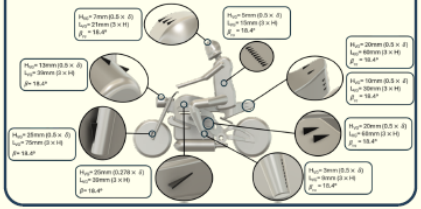
Introduction and Research Gap

As the automotive industry pivots towards sustainability, aerodynamics are crucial for fuel efficiency and emission reduction, primarily by minimising drag. This research aims to fill this critical gap, focusing on improving motorcycle stability and performance through advanced aerodynamic optimisation.

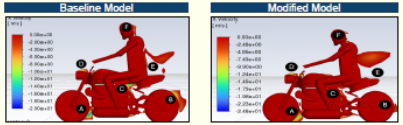
Key Challenges and Research Gap

- Extensive studies on four-wheelers, but motorcycles remain underexplored.
- Motorcycle geometry and rider exposure create unique aerodynamic challenges.

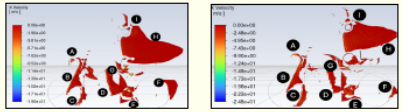
Vortex Generators Placement



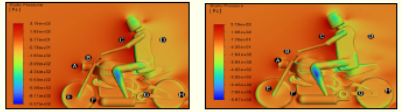
Results and Flow Visualisation



Point A and B, the recirculation bubble is reduced compared to the baseline due to VGs on the front fork re-energising the boundary layer. Point C, VGs near the battery module reduce backflow, leaving only a thin recirculation zone. Similar effect at point F, helmet. Point E the VGs on rider's back makes the flow stronger, but uniform which reduces unsteady wake

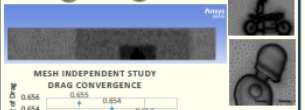


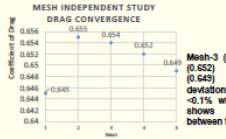
Point A, separation persists but is recirculated downstream, improving uniformity. The VG at the front suspension reduces separation, while a major improvement is observed in the tank and frame. At point D, the rider's leg shows a more uniform flow due to VG integration. VGs on the rider's head improve flow from the rear of the head to the neck and back, while side body VGs modify upstream airflow and reduce concentrated separation zones.



High-pressure stagnation occurs at the point (A), with smoother pressure drops at the top (B) than the baseline. Similar effect at the point (E) and (G), while wakes behind them (F, H) form stronger low-pressure region. Overall, VGs reduce early separation, wakes, and smooth pressure distribution by keeping flow attached and reducing adverse pressure gradients.

Methodology

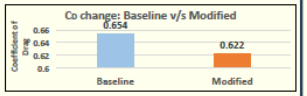




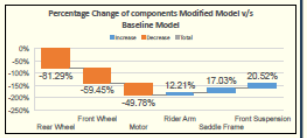
Mesh-3 (0.654), Mesh-4 (0.652) and Mesh-5 (0.645) showed a deviation in C_D values <0.1% which positively shows convergence between the meshes.

INLET: Velocity inlet with a freestream velocity of 22.22m/s
GROUND: Moving with velocity of 22.22m/s
OUTLET: Zero-gauge pressure
TUNNEL WALLS & MOTORCYCLE: Stationary (Zero-slip condition)

Conclusion



The triangular vortex generator reduced the overall C_D by 4.89% (0.654 to 0.622) when applied at H = 0.55.



- Components such as the rear wheel registered the highest reduction (–81.29%), followed by the front suspension (–59.45%) and motor (–49.78%). These components were susceptible to flow separation and, therefore, significant improvement can be observed in the components and showing that VGs have effectively delayed boundary layer separation.
- Whereas some components, such as the front suspension (20.52%), saddle frame (17.03%) and rider arm (12.21%), have shown an increase in total drag. It can indicate that a modification in the airflow caused by the VGs has led to increased aerodynamic drag in these components.

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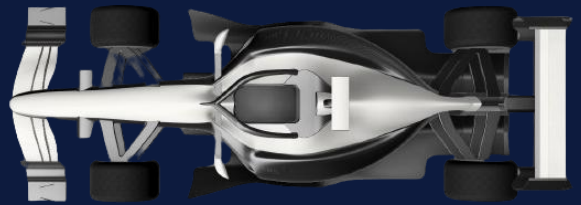
02

FORMULA-1 CARS

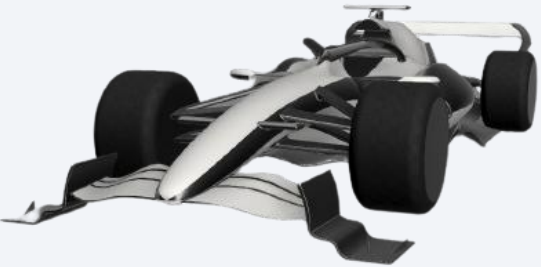
2018 · 2022 · 2026 FIA Technical Regulations

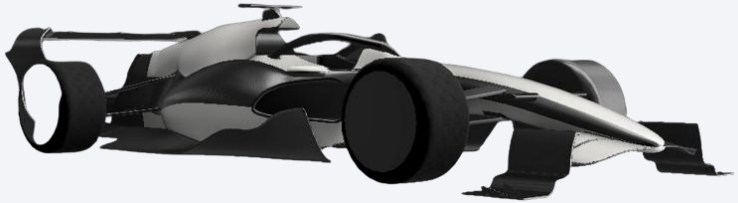


FORMULA-1 CAR 2026 REGULATION

PROJECT OVERVIEW

A concept Formula 1 car developed based on 2026 FIA Technical Regulations, focusing on aerodynamic surface design and flow structure analysis using CFD.





KEY FEATURES & HIGHLIGHTS

1. Parametric surfacing applied to monocoque, sidepods, and floor geometry
2. Structural elements modelled using simplified profiles for conceptual representation
3. Aero surfaces: multi-element front wing, active rear wing concept
4. Followed by full CFD simulation in Ansys Fluent 2024 R2

TECHNICAL SPECIFICATIONS

Regulation Year	2026 FIA Technical Reg.
CAD Software	SOLIDWORKS 2024
Chassis Method	Surfacing + Weldments
Drawings	GD&T (FIA spec)
Aero Package	Front wing, Floor, Rear wing

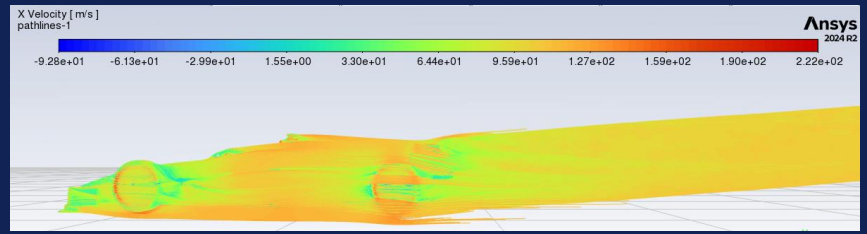
CFD SIMULATION 2026 F1 CAR using ANSYS FLUENT

PROJECT OVERVIEW

Steady-state RANS simulation using k-ω SST model. Velocity inlet at 61 m/s, pressure outlet at 0 Pa, moving ground with rotating wheels. Near-wall treatment with $y^+ \approx 30$

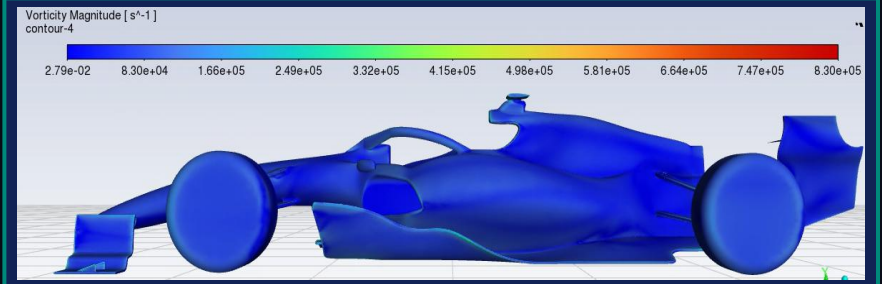
- Model: Steady-state RANS
- Turbulence: k-ω SST
- Freestream velocity: 61.1 m/s (220 km/h)
- Domain: External aerodynamic flow
- Ground condition: (moving floor at 220km/h)

X-Velocity Wake Profile



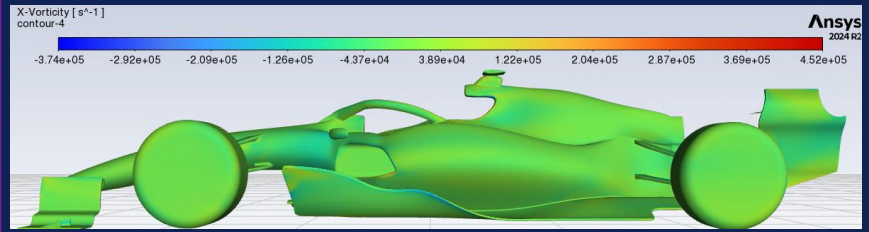
Wake structure visualised using velocity contours to identify low-energy regions

Vorticity Magnitude



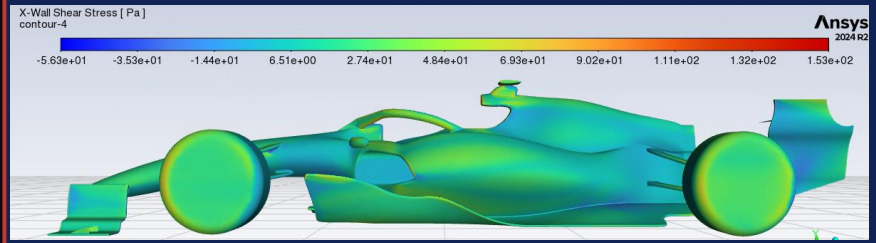
Vorticity magnitude used to analyse vortex generation from front wing and floor edges

X-Vorticity Distribution



Flow structures indicate interaction between front wing outwash and downstream diffuser performance

X-Wall Shear Stress [Pa]

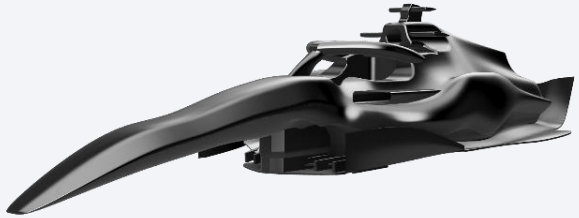


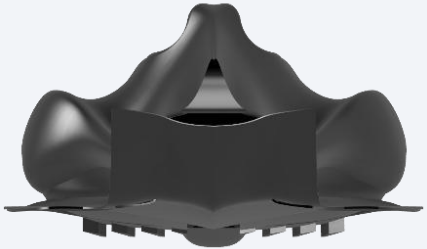
Wall shear stress used to assess boundary layer behaviour

FORMULA-1 CAR 2022 to 2025 REGULATION

PROJECT OVERVIEW

Ground-up F1 design based on the 2022 FIA Technical Regulations, the era that introduced ground-effect aerodynamics back to the sport.





KEY FEATURES & HIGHLIGHTS

1. Ground-effect floor geometry with venturi tunnels and diffuser sections accurately surfaced
2. High-sidepod architecture following 2022 regulation aero philosophy
3. Distinct multi-element front wing with five-plane cascade design
4. GD&T drawings fully annotated to FIA 2022 tolerance specification
5. Rendered in British Racing Green with carbon-fibre texture details

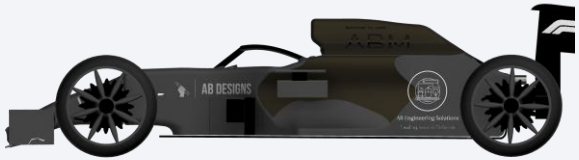
TECHNICAL SPECIFICATIONS

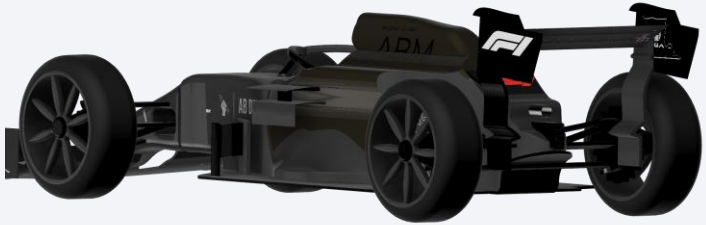
Regulation Year	2022 FIA Technical Reg.
CAD Software	SOLIDWORKS 2024
Aero Era	Ground-effect floor generation
Drawings	GD&T annotated
Key Feature	Multi-element front & rear wings

FORMULA-1 CAR 2018 to 2021 REGULATION

PROJECT OVERVIEW

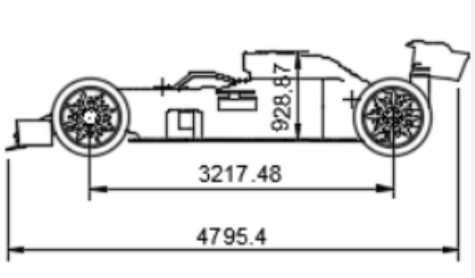
The earliest of three F1 design iterations, modelled to 2018 FIA Technical Regulations. This project pioneered the author's F1 CAD workflow, establishing a repeatable methodology later refined across 2022 and 2026 designs.

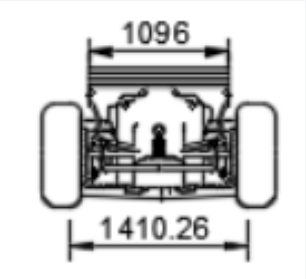


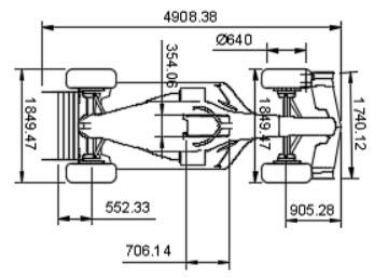


KEY FEATURES & HIGHLIGHTS

1. Aerodynamic surfaces created using Style (ISDX) and Boundary Blend tools, ensuring smooth curvature transitions and surface continuity
2. Structural components represented using simplified parametric geometry to define suspension layout and overall packaging
3. Fully dimensioned 2D drawings including wheelbase 4795 mm, track 1849 mm
4. Assembly constructed using top-down design methodology with datum references and constrained component placement







TECHNICAL SPECIFICATIONS

Regulation Year	2018 FIA Technical Reg.
CAD Software	PTC Creo
Chassis	Parametric Design
Aero Surfaces	ISDX and Boundary Blend tool
Drawings	Full dimension + GD&T

03

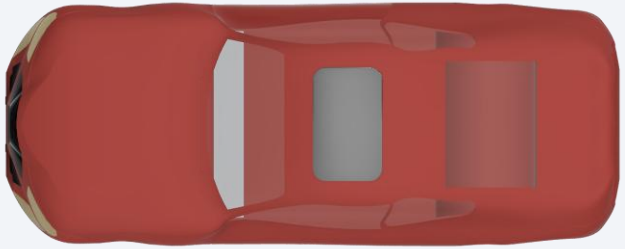
SEDAN & HYPERCAR

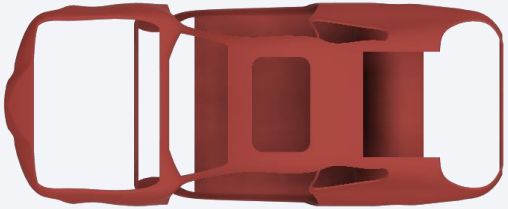
Surface Modelling · FEA · CFD Analysis

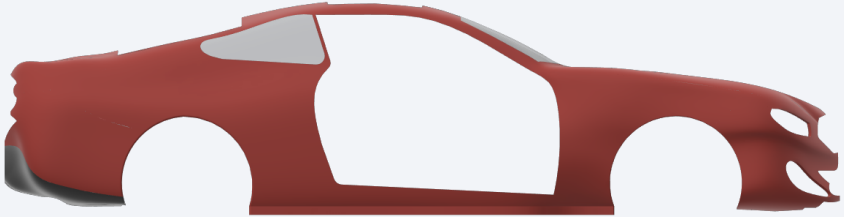
SEDAN CAR

PROJECT OVERVIEW

A sedan car was designed in both SOLIDWORKS and Siemens NX, then subjected to independent FEA and CFD simulation studies in Ansys 2024 R2.







KEY FEATURES & HIGHLIGHTS

1. FEA: Max total deformation 0.11868 mm at highway structural load case (body shell)
2. FEA: Body-in-white mesh with tetrahedral elements and inflation at stress risers
3. CFD: Stagnation pressure 1620 Pa at front; negative pressure 3380 Pa at rear
4. CFD: Turbulent intensity highlights wake separation and underbody diffuser performance
5. CFD: X-velocity contours show boundary-layer development along roofline and boot

TECHNICAL SPECIFICATIONS

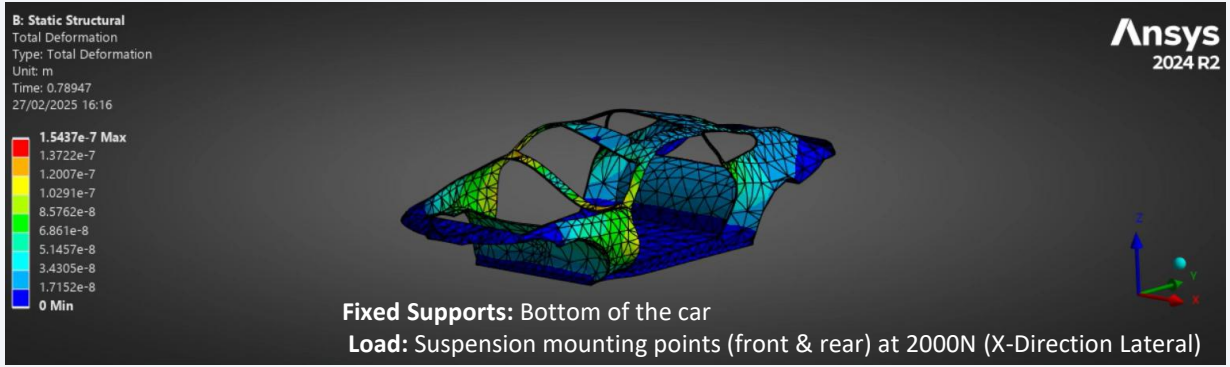
FEA Solver	Ansys Static Structural 2024 R2
FEA Output	Total deformation [m]
CFD Solver	Ansys Fluent 2024 R2
CFD Outputs	Pressure · X-Vel · Turbulence
Mesh Type	Polyhedral + inflation layers
Flow Speed	40 m/s (X-velocity freestream)

SEDAN CAR

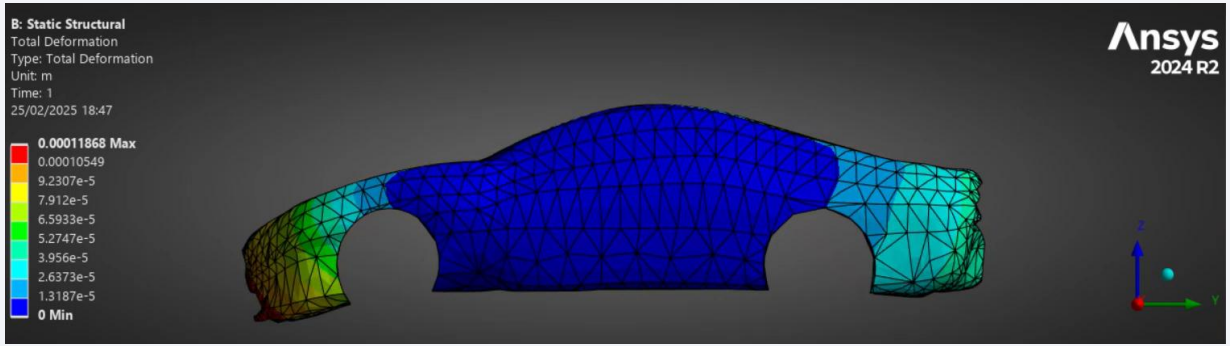
Body-in-White: Structural Steel
•Young's Modulus: 210 GPa
•Poisson's Ratio: 0.3

PROJECT OVERVIEW

A sedan car was designed in both SOLIDWORKS and Siemens NX, then subjected to independent FEA and CFD simulation studies in Ansys 2024 R2.



Fixed Supports: Bottom of the car
Load: Suspension mounting points (front & rear) at 2000N (X-Direction Lateral)



Fixed Supports: Bottom of the car
Load: Suspension mounting points (front & rear) at 2000N (Z-Direction Pothole)

KEY FEATURES & HIGHLIGHTS

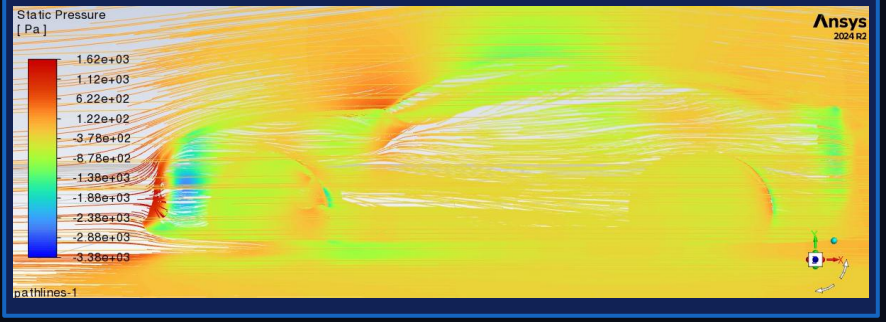
1. FEA: Max total deformation 0.11868 mm at highway structural load case (body shell)
2. FEA: Body-in-white mesh with tetrahedral elements and inflation at stress risers
3. CFD: Stagnation pressure 1620 Pa at front; negative pressure 3380 Pa at rear
4. CFD: Turbulent intensity highlights wake separation and underbody diffuser performance
5. CFD: X-velocity contours show boundary-layer development along roofline and boot

TECHNICAL SPECIFICATIONS

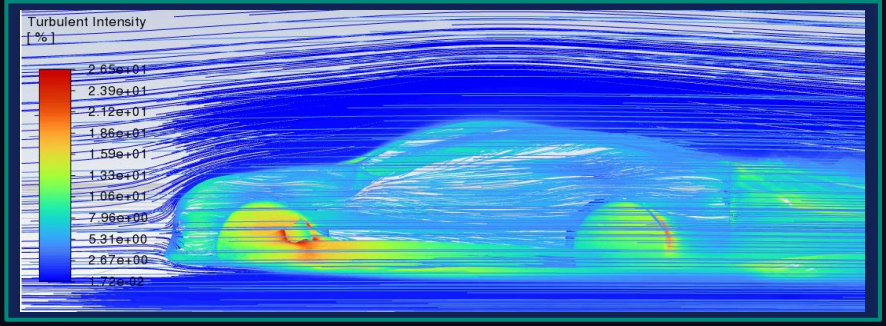
FEA Solver	Ansys Static Structural 2024 R2
FEA Output	Total deformation [m]
CFD Solver	Ansys Fluent 2024 R2
CFD Outputs	Pressure · X-Vel · Turbulence
Mesh Type	Tetrahedral mesh with local refinement and inflation layer
Flow Speed	51.5 m/s (X-velocity freestream)

CFD SIMULATION 2026 F1 CAR using ANSYS FLUENT

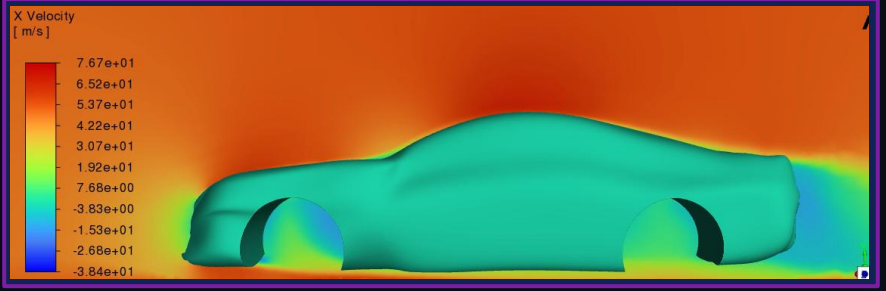
Static Pressure



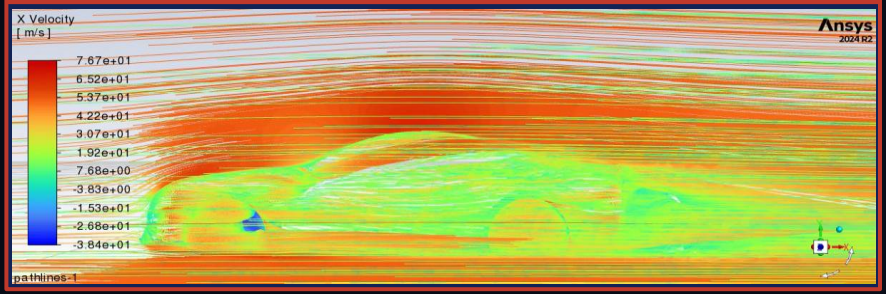
Turbulence Intensity



X-Velocity



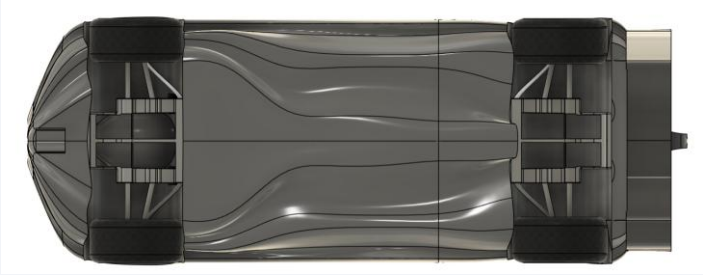
X-Velocity

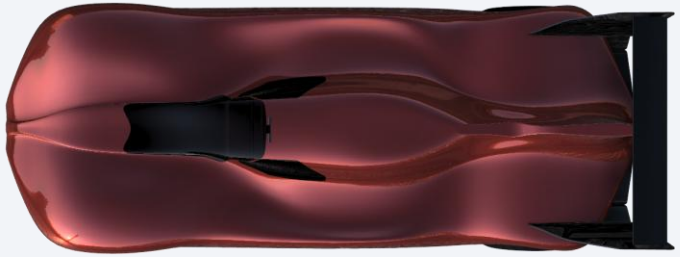


HYPER CAR

PROJECT OVERVIEW

An original hypercar concept designed in Autodesk Fusion 360, exploiting the T-Spline / Form environment for highly organic, sculpture-like body surfaces. The design language draws inspiration from Le Mans prototype racers with low, wide, with dramatic fender arches and an exposed cockpit.





KEY FEATURES & HIGHLIGHTS

1. T-Spline Form modelling enables organic, topology-flexible surfaces beyond NURBS limits
2. Fender arches, cabin, and diffuser sculpted as a single continuous Sub-D body
3. Top-down and side orthographic views confirm proportion and packaging
4. Dramatic deep crimson metallic rendering.
5. Design stage retains full Fusion 360 timeline history for iterative refinement

TECHNICAL SPECIFICATIONS

CAD Platform	Autodesk Fusion 360
Body Method	T-Spline Form (Sub-D modelling)
Design Theme	Le Mans LMP-inspired concept
Surface Type	Organic and continuous G2 surfaces
Render	Fusion 360 Render with HDR studio

THANK YOU

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For full project files, renders, and simulation reports visit LinkedIn